

What is Excel?

Microsoft excel is a spreadsheet software used to store, organize, analyze and visualize data. Excel turn raw data into meaningful information and insights.

Purpose of Excel?

Clean Raw data

Build Interactive dashboards

Visualize data trends.

Manage structured information

Types of data in excel:

Text: Names Categories

Number: sales, quantity

Dates: time based analysis.

Boolean: True (1), False (0)

Geographical data: longitude, latitude

Excel features for Data analytics [23/09/2026]

Data extraction

Data Cleaning

Remove duplicate

Handling the missing values

Removes the irrelevant data and adding necessary data

Text columns, Find and replace.

Formulas and functions:

Basic :- Sum, average, count

logical :- if, and or, not.

lookup: - lookup, xlookup, vlookup, hlookup
Text: - concat, upper, lower.

Data visualization:

Charts: Bar, col, pie... etc
conditional formatting
sparklines

Pivot tables:

Summarize table

Group and filters data

Generate insights quickly

Sorting and filtering

Data validation (Extracting the proper data)

Advantage:

- * Beginner friendly

- * Fast for small to medium dataset

- * Widely used in companies

- * Integrates with tools like power

Disadvantage:

- * Not suitable for large datasets

- * Limited automation compared

- * to coding too.

- * Manual errors are possible

[24/09/2026]

Functions

Set of instruction / set of code is already pre defined is called functions.

There are 3 main components of function

- i) Function name
- ii) Number of inputs
- iii) Return value

Define:

Lower case:

The lower function converts all letters in a text string to lower letters.

Syntax: = LOWER (text)

Example: = LOWER ('ARCHANA') → archana

Define:

Upper case:

The upper function converts all letter in a text string to upper case letters

Syntax: = UPPER (text)

Example: = UPPER ('archana') → ARCHANA

Proper case:

The proper function converts the first letter of each word to upper case and the remaining letters to lower case

Syntax: = PROPER (text)

Example: = PROPER ('Arun') → Arun

Ar.

Len:

The len function counts the number of characters in text, string, including space.

Syntax: = LEN (text)

Example: = LEN ('Data Analyst') → 12

Left function

The left function extract a specified numbers of characters from the beginning (left side) of a text string

Syntax: = LEFT (text, num-chars)

Example: LEFT ("Data analyst", 4) → Data

Right function:

The right function extracts a specified number of characters from the end (right side) of a text string.

Syntax: = RIGHT (text, num, chars)

Example: = RIGHT ("Data analyst", 7) → Analyst.

Mid function:

The mid function extract a specific number of characters from the middle of a text string, starting from a specified position.

Syntax: = MID (text, start-num, num-chars)

Example: = MID ("Data analyst", 6, 7) → Analyst.

Mid function:

The mid function extract a specific number of characters from the middle of a text string, starting from a specified position

Syntax: = MID (text, start-num, num-chars)

Example: = MID ("Data analyst", 6, 7) = analyst.

Text extract using left and Len (First-half)

The left len combination is used to extract the first half of the characters from a text string

Syntax: = LEFT (text, LEN (text) / 2)

Example: = LEFT (A1, LEN (A1) / 2) name: Rahul.

Result: RA.

Second - half characters extraction:

The right len combination is used to extract the last half of a text string.

Syntax: = RIGHT (text, LEN (text) / 2)

Example: If A1 contains RAHUL

= RIGHT (A1, LEN (A1) / 2)

Result: UL

Round-up function:

The round-up function rounds a number upward, away from zero to a specified number of digits.

Syntax: = LEFT (text, ROUND UP (LEN(text)/2, 0))

Example: If A1 contains RAIDUL
= LEFT (A1, ROUND UP (LEN (A1) / 2, 0))

Result: RAH..